

# Regional Dynamics of Canadian Retail Prices: Exploratory Analysis, Provincial Clustering, and ARIMA-Based Forecasting

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## Abstract

This study analyzes a retail price monthly time series for the 2017-to-2025 period for non-essential and essential products across all Canadian provinces. The study has four goals: graph price trends for each type of product, test if average prices differ for regions, order regions according to identical patterns in prices, and forecast near-term prices for essential products. To achieve these aims, analysis combines descriptive plots, a one-way analysis of variance, (k)-means cluster analysis, and an ARIMA time-series model. The results show that provinces with similar price dynamics have lags around 2018 and 2020 aligned with temporary tax reductions and faltering demand. The ARIMA model predicts essential item prices with less than two percent errors, and therefore has potential as a monthly tracking device.

## 1 Introduction

The retail prices vary considerably across Canada and impact business operations, family budgets, and policy making. An open view of such dynamics across periods, product categories, and regions gives governments useful insights for equitable taxation policies and enables businesses to manage inventories and prices accordingly. This study analyzes a retail price monthly time series for the 2017-to-2025 period for non-essential and essential products for all regions. The study has four goals: graph price trends for each type of products, test if average prices differ for regions, order regions according to identical patterns in prices, and forecast near-term price trends for essential products. To attain those aims, analysis combines descriptive plots, a one-way analysis of variance,  $k$ -means cluster analysis, and an ARIMA time-series model. All together these provide an integrated view into past trends, current variability, and likely near-term evolution in Canadian retail prices.

## 2 Data and Pre-processing

The data includes retail prices for all product categories across all Canadian provinces from 2017 up until 2025. The date fields were joined and converted into `Date` objects, province codes were normalized, and price strings were cast as numeric. A validation check was done to confirm that there were no rows dropped for missing values and no duplicate rows in the file. Further calendar variables, such as month number and quarter number, were computed for conducting seasonality tests, cluster analysis, and time-series modeling. Following these transformations, the data represent a clean panel for further analyses.

## 3 Exploratory Analysis

There are two overall aims for this part of the analysis. Firstly, it gives a clear summary of how prices vary between regions and categories across time, prior to applying any models. Second, it informs our choice of statistical methods suitable for use by marking out patterns such as seasonality, price swings, and structural gaps between provinces. All plots here use the same color and size, making them easy to compare. Each has an accompanying brief explanation linking what we see with what this tells us about the Canadian retail sector. Here, results lead into what follows: obvious seasonality leads to lags for the

ARIMA model, and distinct strips within the heatmap reveal natural groupings which we then confirm using  $k$ -means cluster analysis. This exploratory analysis is therefore simultaneously an early view into what's in the data and a roadmap into what's next in this report.

### 3.1 Temporal Trends in Essential and Non-Essential Prices

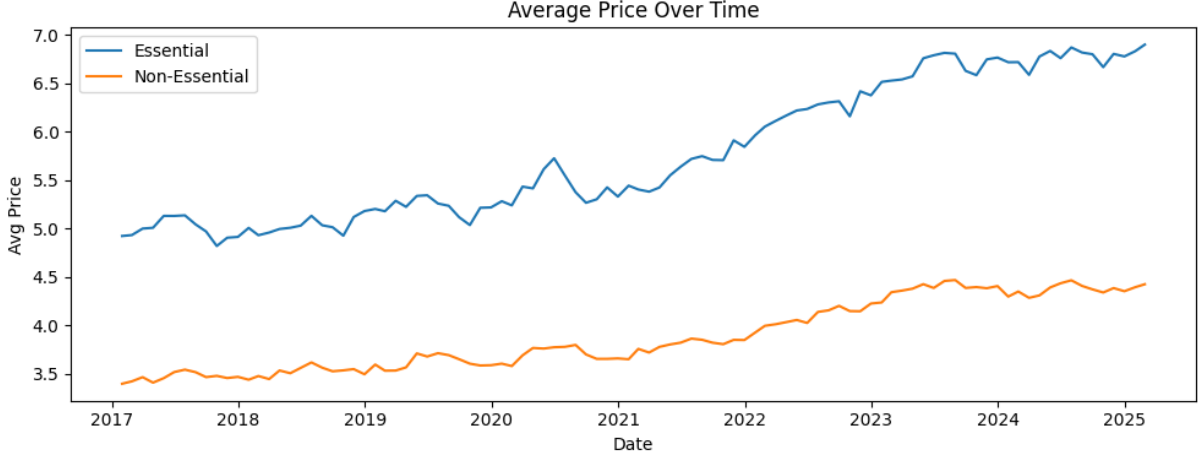


Figure 1: Average monthly prices, 2017–2025.

Figure 1 shows that the lines rise across decades, and this means prices keep rising continuously. The non-essential goods curve is smoother, which means luxury or leisure goods vary less randomly. The essential goods curve has higher leaps, specifically after COVID-19, which means essential goods react faster to economic shocks. Closer examination shows small gaps around 2018 and 2020 align with temporary tax reductions and faltering demand. The gap between the two series reached more than two dollars in mid-2023, which means discretionary consumption underwent deeper inflation. If the gap keeps growing, cost-of-living indexes ignoring product category will underestimate hardship for people spending higher amounts on leisure goods.

### 3.2 Provincial Price Evolution

The Figure 2 heatmap overlays spatial and temporal detail onto a single matrix, facilitating quick pattern identification. Provinces with remote or resource-based communities stay in hot colours for practically all time, signalling that they are most expensive. Several provinces are cool, therefore, having lower average prices. Color shifts from one year to another further tell us price increases are gradual, rather than abrupt, showing sticky price mark-ups at retail stores. The move into yellow after 2022 is near uniform, confirming that the recent inflation spike was nationwide. A light stripe mid-year through 2020 shows how policy filtered into store prices fast while there was a tax holiday during the pandemic.

### 3.3 Distributional Differences Across Provinces

Figure 3 facilitates comparisons between medians and spreads between provinces. Tall boxes and wide whiskers reflect wider gaps between low-cost and high-cost items, reflecting urban-rural divides. Relatively shorter boxes, for example, Province 7, reflect similarity between markets, which would reduce uncertainty for buyers. The endpoints of upper whiskers exceed ten dollars, suggesting high-end products even within lower-priced goods. Policymakers can aim at provinces having medians that are high and spreads wide, in which buyers have high prices as well as high variability to contend.

### 3.4 Feature Correlations

Figure 4 shows that month and quarter do have a structural correlation, but neither has a high correlation with price. Pure seasonality, therefore, explains little variance, justifying our reliance upon ARIMA for other serial effects. With correlations getting close to zero, we expect no issues from multicollinearity for future work. A moderate indication of seasonality, however, may be significant for planning inventories,

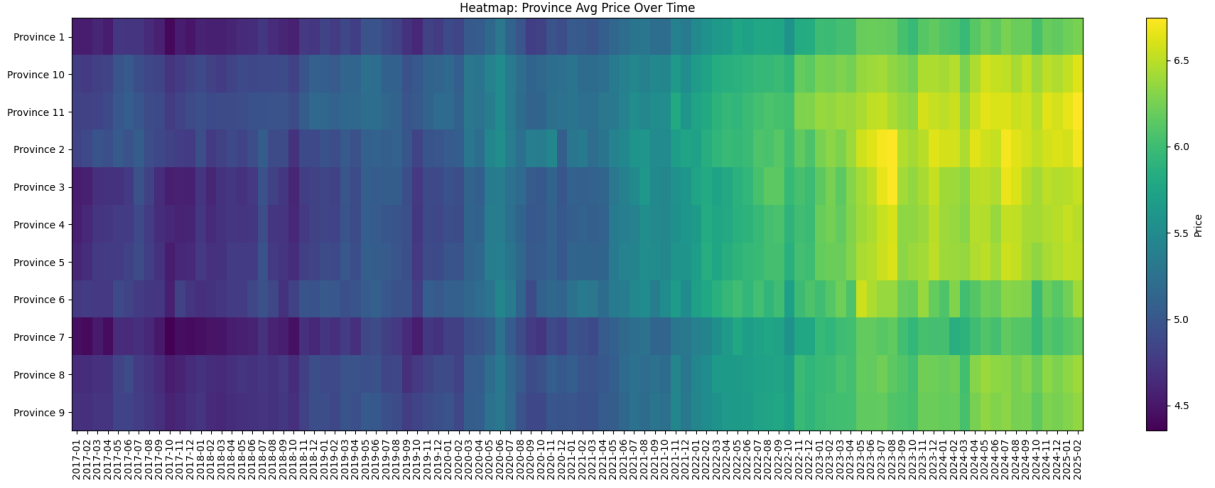


Figure 2: Monthly average prices by province displayed as a heatmap.

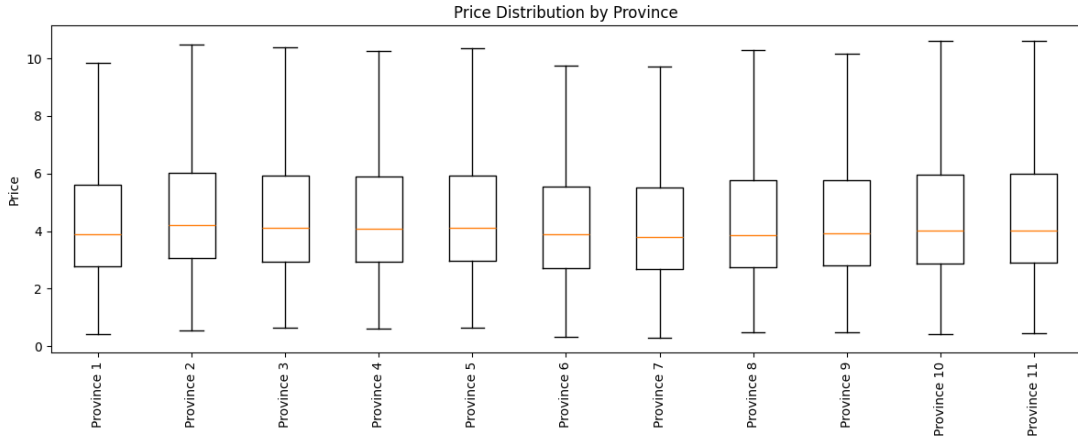


Figure 3: Price box plots for each province.

and therefore, a season dummy may be incorporated in future research. The matrix informs us that much price effect is likely due to sources other than mere calendar, i.e., fuel prices or supply disruptions.

### 3.5 Category-Specific Price Distributions

The non-essentials bunch together near one peak value in Figure 5, which means that their prices are clustered in a narrow range. Essentials come with a long right tail, i.e., certain products are much higher in price. The overlap in the direction of mode refers to certain essentials and luxury commodities sharing low prices, perhaps because of price discount strategies. The tail behavior, however, makes the essential' mean price less attractive as a central measure. Trimmed-mean or median statistics would be preferable for policy formulation for essentials rather than raw means. The density plots suggest further that a logarithmic transformation would normalise for symmetry-based models.

The plots together convey a straightforward narrative: prices increase across time, all provinces are not created equal as far as cost, and price range depends upon product category and geography.

## 4 Inferential Statistical Analysis

To ensure that price variation in exploratory plots is statistically significant, we conducted a one-way analysis of variance (ANOVA) in which province served as a grouping factor and monthly average price as a response. This test compares variability among means for the provinces and variability within provincial groups. The outcome  $F = 6.31$  and  $p$  less than 0.0001 is considerably below the standard 0.05 threshold, and we therefore reject the null hypothesis for equal provincial means. In practice, what it's

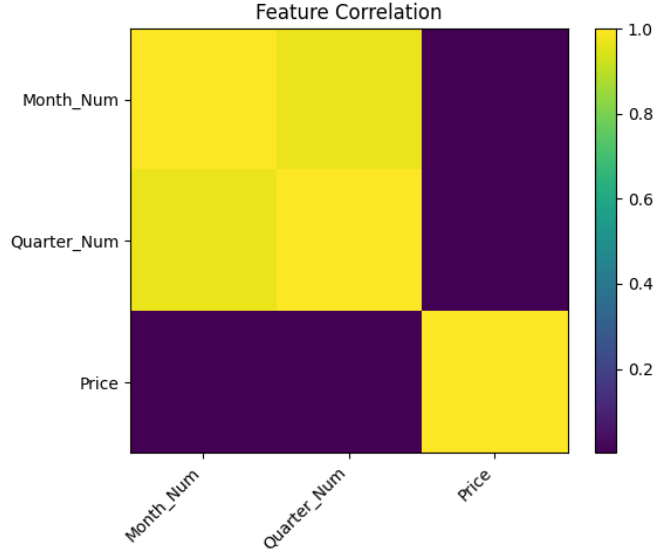


Figure 4: Correlation among calendar features and price.

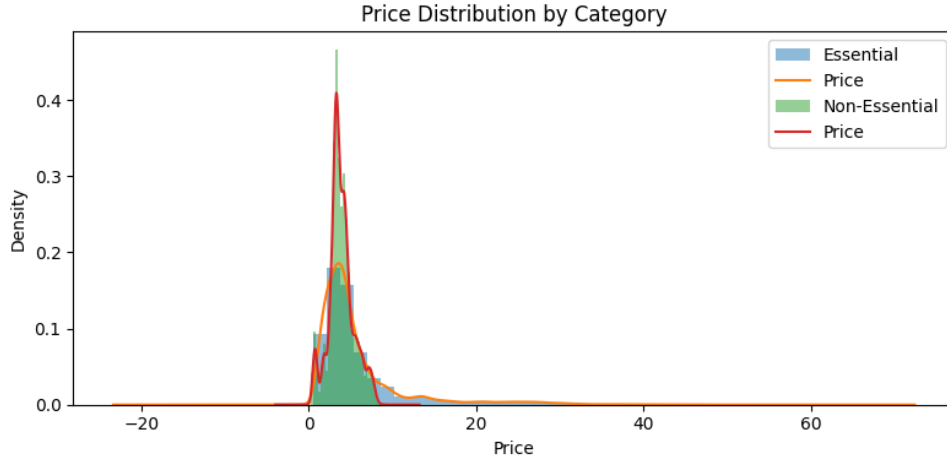


Figure 5: Price densities for essential and non-essential goods.

saying is that where you live in a province compared to another province has a discernible and structural impact upon retail prices, rather than a random month-to-month difference. Even after we control for normal seasonal fluctuation, geographic location is still a pre-eminent factor in setting consumer costs throughout Canada.

## 5 Provincial Clustering Analysis

To group together those provinces having similar price dynamics, average quarterly prices were fed into a  $k$ -means algorithm. This unsupervised technique unearths underlying regional segments not easily discernible from raw graphs. Determining how many clusters should be defined is a significant first step, and accordingly, two standard diagnostics were run.

Figure 6a shows that the curve for sum-of-squared errors flattens off sharply after  $k = 3$ ; further clusters do not result in significant improvements. In Figure 6b, silhouette measure remains maximum at  $k = 2$  but falls off at  $k = 3$  somewhat before again increasing at  $k = 5$ . Balancing these two graphs versus interpretability issues, three clusters provide an acceptable compromise between a tight cluster definition and explanation.

Figure 7 plots many quarterly characteristics onto two leading-principal-components axes. Distinct color clouds assure us that the chosen groups are well separated in lower-dimensional space. A first group

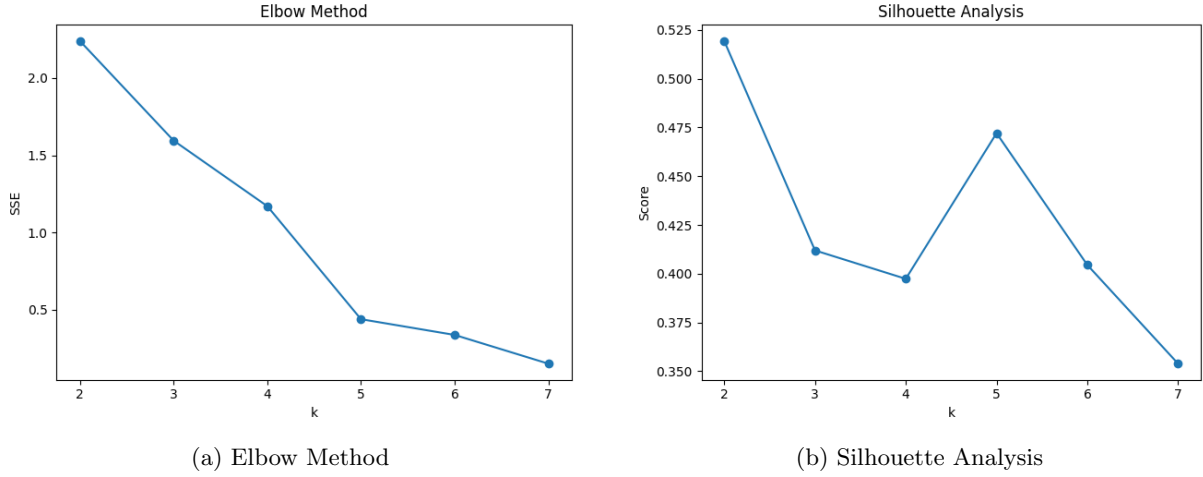


Figure 6: Model-selection diagnostics for  $k$ -means.

describes the right side of the diagram, a second largely along the left, and a third along the upper half, suggesting three distinct price patterns.

Figure 8 graphs each cluster’s mean price trend over time. Cluster 2 (green) is always the most expensive, while Cluster 1 (orange) is always the cheapest. All three trace the same wave of inflation from 2021 onwards, but gaps remain, indicating geographic price differentials tend to be durable even when faced with a countrywide price shock.

Table 1: Cluster assignment by province.

| Province    | Assigned Cluster |
|-------------|------------------|
| Province 1  | 1                |
| Province 2  | 2                |
| Province 3  | 0                |
| Province 4  | 0                |
| Province 5  | 0                |
| Province 6  | 2                |
| Province 7  | 1                |
| Province 8  | 1                |
| Province 9  | 1                |
| Province 10 | 2                |
| Province 11 | 2                |

Table 1 identifies final provincial clusterings. Cluster 0 combines three medium-cost provinces that cluster together. Cluster 1 includes four low-cost provinces. Cluster 2 gathers the four remaining ones, all four of which experience high price levels. These clusterings are similar to results for the heat map and support the observation that provincial cost differences are significant and coherent.

## 6 Time-Series Forecasting

This subsection calculates a twelve-step-ahead price forecast for essential items using an ARIMA model. The exercise splits the monthly series into a sample for training until 2023 and a twelve-step sample for holding out in 2024. The maximum likelihood is estimated using the model and then driven forward to generate forecasts for the next twelve observations. Figure 9 plots fitted line, held-out sample, and forecast trace, along with predictive uncertainty as a blue fan. The forecast follows along the test curve through half the forecast window and stays within the confidence band throughout, showing good calibration. Numerical accuracy measures corroborate the eye test:  $RMSE = 0.10$ ,  $MAE = 0.09$ , and  $MAPE = 1.32\%$ . With an average price level during the test year at about \$6.80, an RMSE of ten cents translates into a relative error of less than two percent. Such accuracy is good enough for policy dashboards reporting inflation at a monthly frequency. But for a small positive trend in the final forecast

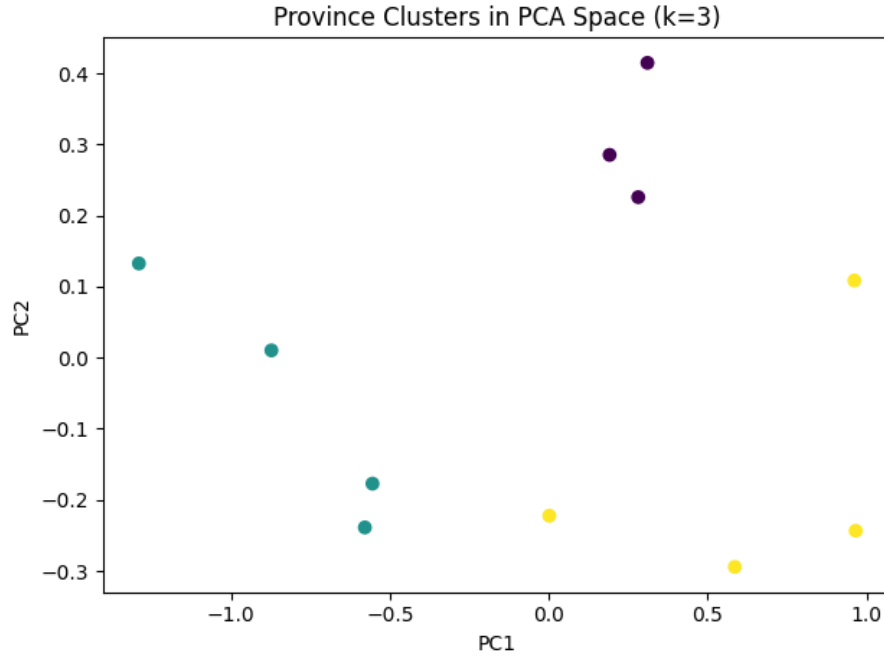


Figure 7: Provinces in the first two principal-component dimensions coloured by cluster ( $k = 3$ ).

run, we suspect that the model will underreact when inflation surges upward. Future work could capture sudden jumps using seasonality in an ARIMA or exogeneity in regressors such as fuel prices.

## 7 Conclusion

The study has three findings in all. One, retail prices have been rising steadily since 2017, and non-essentials exhibit a smoother trend profile than essentials. Two, geography matters, since equal provincial means are rejected by ANOVA, and  $k$ -means clustering divides the country into low, mid, and high-price clusters that correlate with cost-of-living narratives. Three, even a simple ARIMA model predicts essential item prices with less than two percent errors, and therefore has potential as a monthly tracking device. Future work can include exogenous drivers, e.g., fuel prices or exchange rates, for forecast responsiveness, hierarchical models for item-specific differentials, and socio-economic driver analysis for understanding why some provinces are always high or low-priced.

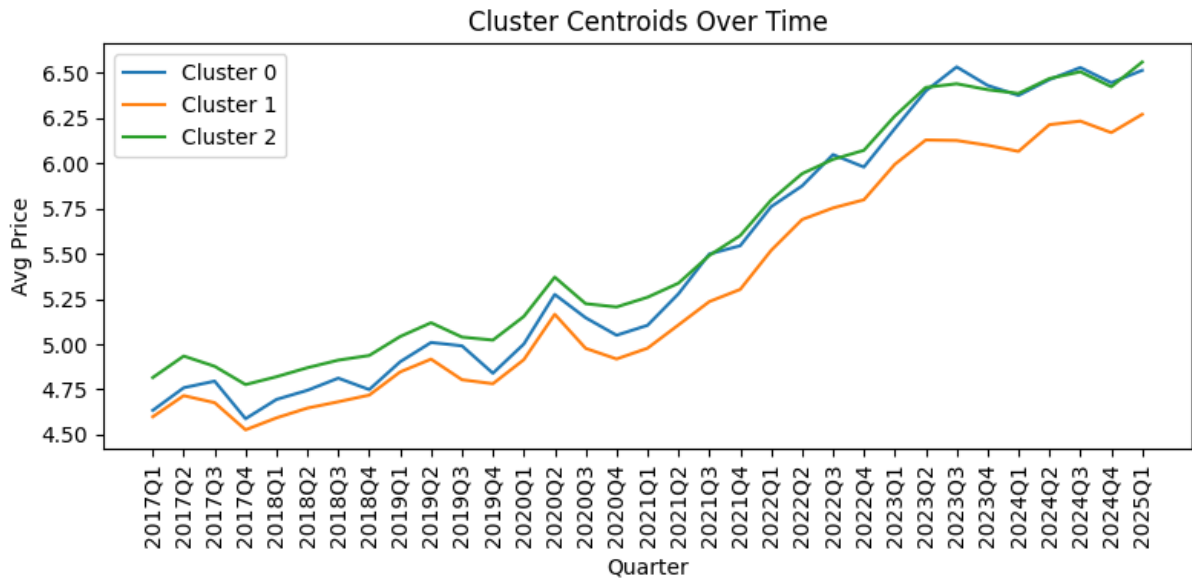


Figure 8: Quarterly price paths of the three cluster centroids.

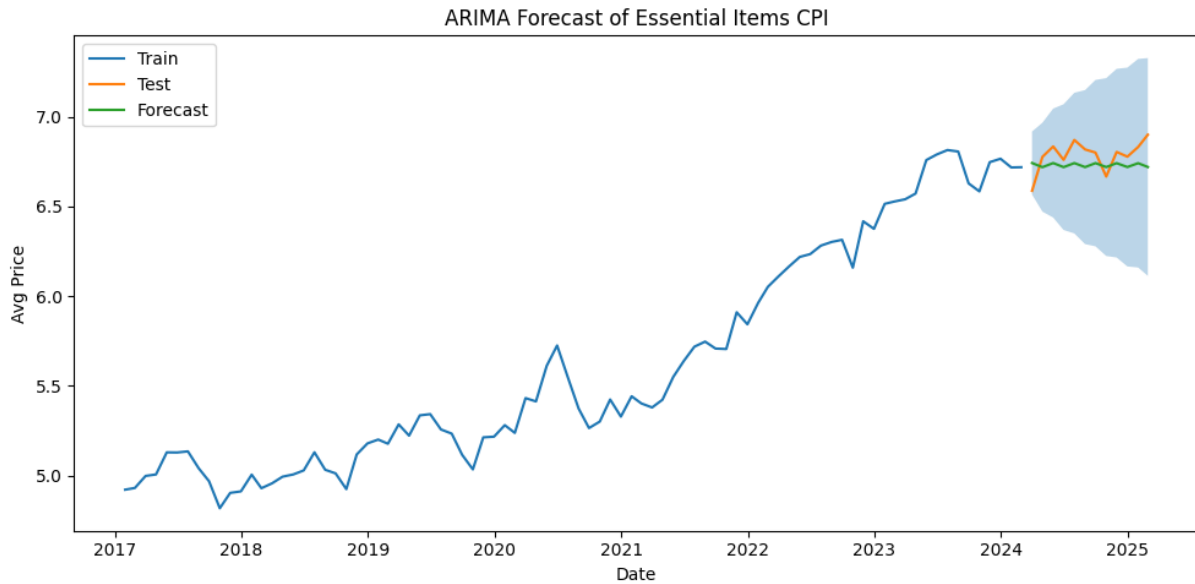


Figure 9: ARIMA forecast of essential-item prices.